Example 1

3×3 Sudoku

The object is to fill all empty squares so that the numbers 1 to 3 each number can only appear once in a row and column.

Step: 1

1	2	3

Step: 2

1	2	3
2	**3**	**1**

Step: 3

1	2	3
2	**3**	**1**
3	**1**	**2**

Try this

How many ways can we arrange the numbers from 1 to 3 in first Row?

[**Answer:** (1 2 3), (1 3 2),(2 1 3),(2 3 1),(3 1 2)]

Example 2

4×4 Sudoku

1	4	3	
3			
	1		3
2		4	1

Answer:

1	4	3	**2**
3	**2**	**1**	**4**
4	1	**2**	3
2	**3**	4	1

Try this

In how many ways can we arrange the numbers from 1 to 4 in first row?

Try this

5, 3, 2 = 15 10 22

9, 2, 4 = 18 36 52

8, 6, 3 = 48 24 66

5, 4, 5 = 20 25 41

therefore 7, 2, 5 = ?

Answer: 14 35 47

Note:

(i) product of 1st 2nd numbers
7 x 2 = 14

(ii) product of 1st 3rd numbers
7 x 5 = 35

(iii) Add these 2 products then subtract the middle number
= 14 + 35 = 49 - 2 = 47

5th Mathematics_Unit_06_Information_EM_Term-1.indd 83 16/02/2023 17:32:31